# Local renormalization group and evolution equations for functional evolution equations in quantum field theories

August 12, 2026

**Abstract**

It has been a long standing open problem in particle and nuclear physics to formulate functional equations for bound states starting from first principle quantum gauge theory. In this paper we discuss such formulations based on functional differential equations. The formulation essentially contains the necessity to fix renormalization group scale in multiscale processes. This problem is addressed using a local renormalization group. We discuss two versions of the local renormalization group. The first one is Wilsonian style. We show that it has a natural formulation using groups of diffeomorphisms. The other formulation uses ideas of resurgence theory. We show that in this formulation renormalization group is related to isomonodromy problem for multidimensional resurgent functions. We discuss evidence that the second formulation can be obtained as a limiting case of the first one and connect this to extremal points in the space of diffeomorphisms. Both formulations are non perturbative.

## 1 Introduction.

It was an original idea of Wilson to consider integration on a subspace of the function space and then, by rescaling the manifold, to bring the path integral back to the original form. This idea was further developed in [1], [2] and [3]. Functional and local versions of the renormalization group were introduced.

It is a basic observation that if the Wilson's partial integration process is applied to the functional $e^{S[f]}$ where $f$ are the fields and $S[f]$ is an action, this simplest exponential form will be modified into an integrand of the form $e^{S_{eff}[g]}G[g]$ where $g$ runs over the new function space, $G[g]$ is some functional of subexponential type and $S_{eff}$ is the effective action.

Important motivation for our work is the Light Cone approach to QCD [4–6]. This approach has the advantages that, first, it provides closed form equations for bound states, second, it explicitly talks about the towers of partonic wave functions. We generalize this approach in two ways. First, we extend it to include scattering events, not only bound states. We achieve this by introducing local functional time. Second, we provide a covariant approach that is not tied to the light cone, and can be formulated in any coordinate system, making it closer to the usual canonical quantization. Third, we formulate our evolution equations for functional complexes. We show that functional

complexes is a necessary element of functionally covariant formulation of QCD. This generalizes the scalar wave functionals considered in [4–6].

Another important motivation for this work were prior works on renormalization group [7–10]. In this approach, the use of currents with singularities on digonals of configuration spaces is made explicit. These papers also discuss the role of hiher algebraic structres in renormalization group. Our approach generalizes these results. We consider evolution equations for complexes defined on functional manifolds. We work in analytic continuation of functionals for complex spacetimes, which allows us to treat more explicitly the singularities of the components of the functionals in these complexified configuration spaces. We relate renormalization group to moves of these singularities in the complex configuration spaces. The parameters for these moves are eventually chosen self consistently and matched to the overall expectation of the energy momentum tensor. This is a key new proposal that we make in this paper.

It is the goal of this paper to bring techniques from holomorphic field theory to this discussion. We apply the Wilsonial LRG idea to the case when the functionals have holomorphic coefficients. Our main results are of two kinds. First, we single out a class of holomorphic manifolds well adapted to the discussion of invariance of functional integral with respect to the LRG transformation. These spaces have infinite topological complexity. We make comparisons with the class of Riemannian surfaces of infinite topological type( we give detailed references in the appropriate section).

Second, we observe the need to consider bundles of infinite type, i.e. bundles where the structure group is an operator algebra. The case that corresponds to the invariance of the functional integral is the case when this structure group is a self absorbing operator algebra. In the case of operators on sequence spaces, the notion of self similarity is well expounded in [11]. In our case, we deal with algebras generated by convolution kernels $K(p, q)$ with regular singularities along algebraic subvarieties, where $p, q$ run over the momentum space.

We discuss applications of our formalism to a number of problems of phenomenological interest. First, we consider the classical problem of radiative correction computation for the hydrogen spectrum. While the problem is classical, the ideas of LRG were not applied to it before. The key observation that we make is that spectral functions, i.e. the lineshapes of energy levels, are functional homotopy invariants. Consistent application of Wilsonian ideas in this context makes it necessary to consider complexes of function bundles on function manifolds. We argue that it is the homotopy invariants of these complexes that have physical meaning.

Then we discuss the use of LRG in purported proton mass computations in continuum. While the full details of such computations are not available( partial results appear in a companion paper by the present author, which is currently in press), there are a number of ideas that we already can discuss in our present formalism. The proton has a continuum of energy scales. To fully fix this continuum we need a function of space variables. We propose to identify this function with the Local Renormalization Group field. The full formalism for bound states involves functional versions of mixed Riemann Hilbert factorization problems. This suggests that the LRG field is related to the factorization data in the relevant functional Riemann Hilbert problem.

# 2 Functional integration and Wilsonian Renormalization Group.

The functional integration over a function space is ill defined. One of the reasons for this is that there exists an indefinitely large variety of function subspaces of $C([0,1])$ and similar naturally defined function spaces, such as the space of differentiable functions on a smooth manifold. The set of non isomorphic subspaces is analytic non Borel. In that sense it is strictly unclassifiable [12]. If there were a well defined notion of volume form on function space, it must have been sensitive to the particular type of the Banach space under consideration. In principle, such development is possible to carry out to the end, as we will show elsewhere. For the purpose of this publication, it is sufficient to note that integration with Gaussian weight is well defined on Banach spaces [13]. We will consider functionals of a very special form

$$Z\{S\} = \int Df \; E[S_1[f], ..., S_n[f], ....] \tag{2.1}$$

where

$$S_n[f] = \int (dx) S_{a,\alpha_i}(x) \prod \partial^{\alpha_i} f(x) \tag{2.2}$$

are local functionals of polynomial type and $E[s_1, ...., s_n, ...]$ are functions on $\mathbb{C}^\omega$ that have exponential behavior near infinity in $s_i$ and do not have any other singularities. The requirement that $E$ has exponential behavior means that it satisfies the following asymptotic condition

$$E[s_1, ..., s_i, ...] \sim exp(A_i[\hat{s}]s_i) \sum \frac{B_{i,n}[\hat{s}]}{s_i^n} \tag{2.3}$$

where $\hat{s} = (s_1, ..., s_{i-1}, s_{i+1}, ...)$.

The chain of integration in functional space can then approach infinity as long as it stays in the decaying Stokes sector for the function $E$. Note that we are back to the case of well defined functional integration, as essentially we have a polynomial exponential prefactor of the functional integral.

This type of functionals is motivated by three factors:

1) it is stable with respect to the LRG transformation that we consider in this paper.

2) it is an infinite dimensional generalization of the class of functions considered in irregular Riemann Hilbert correspondence.

3) is is an infinite dimensional generalization of the functions that satisfy flat connections with irregular singularities.

# 3 Compensating Diffeomorphisms.

In this section we consider the most basic case of functional integration and identify the notion of compensating diffeomorphism that underlies the Wilsonian approach to LRG.

We first note that functional integration is naturally a functional of the 3rd kind, i.e. a map

$$\int Df : C(C(M, \mathbb{R}), \mathbb{R}) \to \mathbb{R} \tag{3.4}$$

In the exponential case

$$Z\{S\} = \int Dfexp(S[f]) \tag{3.5}$$

Note that the space of integrable functionals is not separable. Various set theoretical issues [14, 15] play a role in rigorous self consistent definition of functional integral. We will restrict our attention to integrals of exponential type

$$Z\{F\} = \int Dfexp(\int A(x)f(x)^2)F[f] \tag{3.6}$$

where $F[f] = \sum F_n(x_1, ..., x_n)f(x_1)...f(x_n)(dx)$. In this case, measure theory is well known [13]. We assume the function $A(x)$ to be continuous.

Discrete Local Renormalization group transformation consists in the following. The manifold $M$ is split according to

$$M = X \cup Y \tag{3.7}$$

where $X, Y$ have a common boundary $B$. We will denote this split by $s$. In a sense, this split is a choice of submanifold $N \subset M$ of codimension 1. In contrast to topological theories [16], the function values on the boundary do not play a role. This is a result of our assumption of continuity of $A(x)$. In a sense, this is a natural result because intuitively the space of functions defined only on the boundary embeds a infinite codimension subspace into the space of functions defined on $M$, i.e. (any of the infinite variety) embedding $C(B) \to C(M)$ has infinite codimension. If the measure is in some sense nonsingular, we can neglect the value of the function on the boundary

$$\int_{C(M)} Df = \int_{C(X)} Df_1 \int_{C(Y)} Df_2 \tag{3.8}$$

$$Z\{F\} = \int Df_1 \int Df_2 exp(\int_X A_1(x)f_1^2(x) + \int_Y A_2(x)f_2^2(x))F[f_1 + f_2] \tag{3.9}$$

The integration over $f_2$ can be performed as

$$\begin{aligned} Z\{F\} = \int Df_1 exp(\int_X A_1(x)f_1^2(x))(det(A_2))^\alpha \sum C(l)\times \\ \times F_{2n+2l}(x_1, x_1, ..., x_n, x_n, y_1, y_1, ..., y_l, y_l)\times \\ \times f_1^2(x_1)...f^2(x_n)A_2^{-1}(y_1)...A_2^{-1}(y_l)dy_1...dy_l \end{aligned} \tag{3.10}$$

where $C(l)$ are certain constants, and $\alpha$ depends on statistics of $f$.

We perform now the crucial step of transforming the manifold with boundary $X$ back to the original manifold with boundary $M$

$$d : M \to X \tag{3.11}$$

$d$ is the compensating diffeomorphism. This diffeomorphism depends on the split. In coordinates

$$d = d_i(x_j)[s] \tag{3.12}$$

where we wrote the functional variable $s$ explicitly.

We also need to perform the change of functional variable as

$$g(x) = f_1(d(x))\sqrt{\frac{A_1(d(x))}{A(x)}} \tag{3.13}$$

Function $g$ is defined on $M$. Then we have

$$Z\{F\} = \int DgJ[g]exp(\int A(x)g(x)^2)(det(A_2))^{\alpha}\sum C(l)\prod\sqrt{\frac{A_1(d(x_i)}{A(x_i)}}\times \tag{3.14}$$

$$\times F_{2n+2l}(d(x_1), d(x_1), ..., d(x_n), y_1, y_1, ..., y_l, y_l)g^2(x_1)...g^2(x_n)A_2^{-1}(y_1)...A_2^{-1}(y_l)dy_1...dy_l$$

The integrand in this formula is identified with the finite LRG transformation of $F$ under the split $M = X \cup Y$ and the compensating diffeomorphism $d : X \to M$. Taking $Y$ to be an infinitesimally small collar over the boundary of $M$, one can work out the infinitesimal( differential) LRG transformation.

LRG in this case is the following statement:

*For a family of splits $S$( choices of embeddings of a codimension 1 manifold $N \subset M$), there exists a diffeomorphism $d_i(x)[s]$ such that the transformation 3.14 leaves $F[f]$ invariant.*

Wilsonian RG is therefore formulated as existence of a functional

$$d : Emb(N, M) \to Diff(M) \tag{3.15}$$

such that $F$ is left invariant under the action of this functional given by 3.14.

Infinitesimal version corresponds to $M$ having a boundary $\partial M$ and a choice of a function space of sections $N(\partial M)$ of the normal bundle on $\partial M$. The LRG version then is the formula for the variational derivative $\delta/\delta n_i(\vec{\xi})$, where $\vec{\xi} \in \partial M$, and $n_i$ runs over a subspace $N(\partial M)$. This subspace may be smaller than $N(\partial M)$, in particular may have infinite codimension.

# 4 Renormalized QED

The motivation for this section comes from modern calculations of Lamb shift at 2 loops and beyond [17–22]. These papers claim to be consistent with experiment [23,24], but the precision for 1s-2s experiments is larger than theory calculation precision. Proton radius is also not fully accounted in these calculations. Furthermore, on the experimental side, there are potential issues with long tails in the line shapes of energy levels [23]. As we demonstrated [25], these shapes are not Lorenzian and have higher resurgent corrections.

But more importantly, the calculations in the above literature do not fully account for gauge inavairance of multi fermion amplitudes. Gauge invariance in n-particle fermion systems is a well known problem. We propose a solution for this problem. The solution enatils upgrading generating functional into a family of functionals that have dependence on both spacetime variables and on fields:

$$F : M^n \times \{A\} \times \{\psi\} \to V \ . \tag{4.16}$$

In coordinates, this corresponds to considertion of functionals of type

$F_{\alpha_1,...;\mu_1,...;\beta_1,...}(x_1,...;z_1,...;y_1,...)[\bar{\eta},J,\eta]$ where indices $\alpha_i$ correspond to i'th coordinate $x_i$ and so forth. There are compelling mathematical resons to consider such objects. They correspond to forms on the function manifold $(\bar{\eta},J,\eta)$ of fields. In this way, we are generalizing conventional partition function. The precise statement, as we will see, is that these functionals, satisfying appropriate field equations, can be represented as functional Fourier Integral Operators. However, before getting to this result, we need to address the issue of renormalization. As we saw above, renormalization introduces both metric and diffeomorphism degrees of freedom. This expands the space of definition of functionals quite substantially as we will see next.

In the previous sections we discussed renormalization and partial integration over a subspace of functions on a manifold. After such integration, we had to perform rescaling, or transformation back to the original manifold. This has the following implications. While we started with a flat spacetime, after the transformation the spacetime is no longer flat. Therefore, the wave functionals become functionals of metric, in addition to fields:

$$F^{QED} = F[g,A] \tag{4.17}$$

where $g \in Riem^{\mathbb{L}}(M)$ is the space of allowed metric types on our manifold $M^{\mathbb{L}_0}$ with smooth structure $\mathbb{L}_0$, and $A$ is the set of gauge fields. We allow for generalized theories where higher tensor fields $A_{\mu_1,...,\mu_n}(x)$ are included. We also allow for inclusion of spinor fields. This means that we have to pass from metric to tetrad $e^a_\mu(x)$. We are going to omit these subtleties, to keep our notation light. Our main object of interest is the functionals $F_{\alpha_1,...,\alpha_n}(x_1,...,x_n)[g,A]$ defined on $Riem(M) \times YM(M) \times Spin(M)$ defined on the space of metric $Riem(M)$ and field spaces.

These functionals generalize both generating functionals of a QFT and wave functions. These functionals behave as wave functions in variables $x_i$ and indices $\alpha$, while they behave as generating functionals in the functional variables $g, A$, which play the analogy of currents in the partition function. They satisfy functional differential equations that involve operator of the type $\hat{\hat{L}}(x_i, \frac{\partial}{\partial x_i})[A, \frac{\delta}{\delta A}]$.

The considerations in the previous section lead us to consider a broader definition space of functionals. We saw that there are also diffeomorphism variables $f \in Diff^{\mathbb{L}}(M)$. We therefore upgrade our functionals to the following form

$$F^{QED} = F[f,g,A] \tag{4.18}$$

where $f = f_\mu(x_\nu) \in Diff(M)$, $g = g_{\mu\nu}(x) \in Riem(M)$, and $A$ is the set of renormalized multifields. These functionals act as

$$F : Diff(M) \times Riem(M) \times YM(M) \to V. \tag{4.19}$$

for some Frechet space $V$. We define them as sections of the Frechet bundle on submanifolds of the space $Diff(M) \times Riem \times YM(M)$. The Frechet space $V$ is usually infinite dimensional and often looks like $C(Conf_n(M))$ taken to some power.

We would like to note two points of contact of this theory with existing literature. In [26] there is developed formalism for the study of a notion of tangent bundle to spaces of maps between manifolds that include spaces of type $Map(N,M)$ , $Emb(N,M)$ , $Imm(N,M)$ , $Diff(M)$ and subspaces thereof. The subspaces to which this formalism is most applicable include geometrically defined spaces. The simplest example is the space of loops that pass through a specific submanifold at a specific time. This line of

research leads to analogy between the tangent bundles and spectra. To our purposes, this is too restrictive an approach, because passing to topological equivalence forgets part of the frechet structure which is important for formulation of functional partial differential equations.

More close to our approach is the approach in [27, 28] where a notion of spinor bundle over loop spaces is defined. In this approach, explicit $C^*$ bundles are considered. Moreover, a rigging on Hilbert spaces is considered. This is indeed quite close to some of our analytic constructions because the fibers in the functional fiber bundle must include the rigging, in order to account for asymptotic states. We know that plane waves in QFT is an approximation. Rigging makes this precise. We will return to this point in the section on manifolds with cylindrical ends.

There is a remarkable relationship with the literature on diffeomorphism dynamics [29, 30]. In this literature, the main object of study is dynamics of individual diffeomorphisms as well as dynamics of families of diffeomorphisms. Renormalization shows up as the operation $f \to \phi o f^p o \phi^{-1}$ for some simple choice of $\phi$. One of the most interesting objects in this area is the Newhouse domain. Our theory of functionals is directly applicable to the study of this subset in $Diff(M)$. The functional forms $F(x_1, ..., x_n)$ are well defined on open parts of the Newhouse domain. However in order to extract dynamical information on these sets, we have to consider differently defined functional differential operators that, e.g., are more directly connected to iterations of diffeomorphisms.

There is one more level of refinement of the theory. We can go one step deeper and consider smoothing of non smooth topological manifolds $M^{top}$, in the sense of smoothing theory [31]. This is especially relevant in the case of 4-manifolds, in particular exotic $\mathbb{R}^4$ [32]. In other words, we upgrade our functional to

$$F^{QED} = F[s, f, g, A] \tag{4.20}$$

where the functional variable $s$ is the smoothing operation. Note that derivatives are only allowed on the composite objects like $f(s(x))$ after smoothing is already applied. At the level of $M^{top}$ derivatives may exist only on open sets. In this upgrade QED becomes the theory of functional bundles

$$F : Smooth(M) \times Diff(M) \times Riem(M) \times YM(M) \to V \tag{4.21}$$

and functional manifolds constructed by gluing of open balls modeled on the function manifold in this equation. We note that functional differential operators only exist on the level of $f, g, A$, while on the level of $s$ the logic of the equations for $F$ is entirely different. Parts of the relevant theory is discussed in [33–35].

Example covariant equations include the following

$$\frac{\delta}{\delta g_{\mu\nu}(x)} F = \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(x)} F \tag{4.22}$$

and

$$\frac{\delta}{\delta g_{\mu\nu}(x)} F - g^{\sigma\rho}(x) \frac{\delta}{\delta \partial_\mu f_\sigma(x)} \frac{\delta}{\delta \partial_\nu f_\rho(x)} F = 0 \tag{4.23}$$

The logic of LRG action is as follows. We have a diffeomorphism type $M$. Then we define quantized dynamics of gauge fields $A$ and metric $g$ on $M$. This dynamics, described by functional forms $F[g, A]$ turn out to be divergent. We regularize and

eventually renormalize it by applying a diffeomorphism $f$ to manifold $M$, metric $g$, and field $A$. The functional $f$ is dependent on the process, i.e. on the functional $F$. Renormalization is then the fact that $F$ is a fixed point of this process:

$$F[g[f], A[f]] = F[g, A] \tag{4.24}$$

where in this equation $f = \Phi\{F\}$ is the renormalization group field.

The central problem in the above development is th interpretation of the running coupling $\alpha(log(\mu/m_e(log(\mu/...))))$ in terms of function space geometry. In dimension 4, there is such interpretation related to topological structures on $\mathbb{R}^4$. It is related to natural logarithmic structures in Bing involution, as we detail in later sections.

# 5 Laplacian equation on function manifolds and its approximate solutions.

In this section, we collect results on the Laplacian equation on a manifold of gauge fields, up to gauge transformation $\mathbb{M} = \{A\}/\sim$. We put metric on this manifold as follows

$$ds^2 = \int_x \int_y G_{\mu,\nu}(x,y)[A]\delta A_\mu(x)\delta A_\nu(y). \tag{5.25}$$

In general, we can also add the terms

$$ds^2 = \int_x \int_y G_{\mu,\nu}(x,y)[A]\delta A_\mu(x)\delta A_\nu(y) + \sum_\alpha \int_x G_{\mu,\nu,\alpha}(x)[A]\delta A_\mu(x)\delta\partial^\alpha A_\nu(x). \tag{5.26}$$

Of particular interest is the metric

$$ds^2 = \int_x \delta A_\lambda(x)(D(x)[A])^2\delta A_\lambda(x) \tag{5.27}$$

which is gauge invariant but has nontrivial isotropic subspace.

We are going to consider functional Laplacian( an example of functional PDE) for this metric. It is formulated for functional tensors $F_{\alpha_1,...,\alpha_n}(x_1,...,x_n)[A]$, here $\alpha = (\nu_1,...,\nu_s)$ are multi indices. Laplacian is formulated for the following connection

$$\nabla_z = \frac{\delta}{\delta A_z} - \Gamma^z_{x,y}. \tag{5.28}$$

The functional Christoffel symbol is defined as

$$\Gamma_{xy,z} = \delta_x G_{yz} + \delta_y G_{xz} - \delta_z G_{xy}. \tag{5.29}$$

Indices are raised using the contravariant metric $G^{xy}$ which is defined using the integral equation

$$\int_z G^{xz}[A]G_{zy}[A] = \delta^x_y \tag{5.30}$$

In particular, for the metric 5.27, the contravariant metric is the propagator in the external field.

The functional Laplacian has the form

$$\Delta F = EF \tag{5.31}$$

where

$$\Delta F = G^{uv}\nabla_u\nabla_v F. \tag{5.32}$$

Note that the terms $\Gamma$ correspond to 3-gluon vertices, while terms $\Gamma * \Gamma$ correspond to 4-gluon vertices. This can be seen from the structure of $\Gamma_{xy,z}$ that precisely has the right derivative structure of the 3-gluon vertex. For example

$$\delta G^{a,b}_{\mu,\lambda}(x,z)[A]/\delta A^c_\nu(y) = \delta_{\mu,\lambda}\delta^{a,b}D^c_\nu(y)[A]\delta(x,y)\delta(y,z). \tag{5.33}$$

We get the 4-gluon vertex without derivatives if we arrange the propagator in $\Gamma^s_{xu}\Gamma^v_{sy}$ to be after the derivatives in $\Gamma$. This is the familiar ordering issue from deformation quantization.

We now study approximate solutions for this equation. They have the form

$$F \sim exp(-G_{uv}\delta A_u \delta A_v)P[A] \tag{5.34}$$

where $P[A]$ is a polynomial. Note that the manifold of gauge fields gets a natural pseudo compactification due to the exponential factor. We interpret this as a Stokes phenomenon on complexification of $\mathbb{M}$. It is interesting to compare this to the quantization conditions proposed in [6]. Here we are talking about the multidimensional Stokes phenomenon which is a generalization of [36].

Remark: the usual curvature equation now has the form

$$R_{\mu\nu}(x,y)[A]\{G\} = 0 \tag{5.35}$$

# 6 Interpretation of scattering experiments in terms of geometry of function manifolds.

We interpret the solutions to the equation in the previous section by identifying its solutions with stable particles. Then for the scattering problems we get the following equation

$$d\sigma = \int < H_1[A] *_{\vec{Q}} H_2[A]|\hat{\hat{O}}[A, \frac{\delta}{\delta A}| \oplus_i F_i[A] > DA \tag{6.36}$$

where $H_a$ are the functionals corresponding to the particles in the initial state, and $F_i$ are functionals that correspond to the final state.

This equation incorporates necessary asymmetry of particle collisions. On the parton level, there are interactions of arbitrary arity $n \to m$. At the same time, at the level of stable particles, in vacuum, the interactions are of type $2 \to n$. Our formalism explicitly accounts for this asymmetry.

The above equations reduce to the usual perturbative QCD in the limit of Gaussian approximation 5.34. At weak field, we can approximate $\Delta = \int_x \int_y G^{xy}\frac{\delta}{\delta A_x}\frac{\delta}{\delta A_y} + O(A)$ where $O(A)$ contain nonlinear terms in $A$, in particular the terms that correspond to $\Gamma$ and $\Gamma * \Gamma$ and thus account for 3- and 4-gluon vertices. The polynomials $P[A]$ in 5.34 correspond to the polarization vectors $\prod_i \epsilon(s_i, \vec{k_i})$ of the free gluons. The integration over $A$ in 6.36 produces propagators $G^{xy}$ between nonlinear terms that result in the formal expansion in $\Gamma$ and $\Gamma * \Gamma$, if we treat these terms as perturbations.

# 7 LRG and open complex manifolds.

In this section we discuss an interplay between LRG and complex structure on the underlying spaces. Compex structure plays an important role in perturbation theory. Feynman's epsilon prescription makes perturbative correlation functions into holomorphic functions with regular singularities. This structure is exploited heavily in modern study of amplitudes. The algebraic structure of the singularities plays a key role in this theory. Perurbative expansion gives rise to bundles over a complement to a set of varieties $L_I$, the Landau varieties [37]. This set is recursive. Higher order varieties are obtained as a result of application of elimination operations on lower varieties. The result of this is a profinite space with infinitely many ends. In perturbation theory, the ends are algebraic. This is not the case beyond perturbation theory. On general grounds, we expect these ends to be given by entire functions of exponential type, e.g. [38]. The position of these ends in practical theories is the result of the solution of self consistency fixed point equations. Their geometry is an important source of information on non perturbative behaviour of the theory.

We found that there is an actively developing area where similar objects are studied, namely the theory or Riemannian surfaces of infinite genus. The discussion of the emergence of such objects in foliation theory can be found in [39,40]. The Riemannian surface $R_\infty$ of infinite genus is roughly a profinite space obtained by an infinite version of pair-of-pants decomposition. In simple cases, one obtains spaces with finitely many ends [41]. Very abundant class of surfaces is furnished by Cantor bouquet examples [42, 43]. The ends can be accumulated by the genus [43], in which case we have non planar ends. Dehn twist in infinite case was studied in [44]. Mapping class group in infinite case was studied in many publications including [45]. The notion of non displaceable surfaces was discussed in [46].

The two prominent examples of self similar Riemannian surfaces are 2-ended surfaces and surfaces of Cantor type. The 2-ended surface is obtained by gluing a countable infinity of spheres with 4 punctures. This space is evidently self similar. There is a countable infinity of self similarity transformations that consist of cutting at the joint $i$ and then rescaling the remaining space such that this puncture at $i$ goes to infinity.

We note that pair-of-pants decomposition exists also in the case of non singular projective hypersurfaces [47]. In this case, the "pants" is the complement to a set of hyperplanes.

**Def:** Self similar manifolds of type I are profinite manifolds obtained by a countable or continuum iteration of gluing of the standard pair of pants procedure.

The functional integrals that are interesting in this case consist of Gaussians of expressions of the form $\int_{x,y} \bar{f}_a(x) A_{a,b}(x,y) f_b(x)$ where $f_a(x)$ is a section of a bundle on the profinite space $X_\infty$. Note that analogies of the classical bundle theory [48] are interesting in this case, after we endow the bundle $f_a(x)$ with suitable growth condition near the ends. Note the relation with the examples of functions of intermediate exponential growth [49]. Note the relation with Cayley graphs of groups with subexponential growth [50].

The functions we encounter in perturbation theory have singularities on ends of profinite spaces. This motivates

**Def:** Self similar manifolds of type II are complements to manifolds of type I such that self similarity transformation exists. This self similarity transform can permute the self similar components of the end of the manifold.

Note that this notion is already interesting in the 2d case, in connection with compactifications of $\mathbb{C}^2$ [51]. The profinite complexes constructed in [49] can be used to construct examples of compactifications of $\mathbb{C}^2$ by Reimannian surfaces of exponential type.

In the case of type II manifolds, the interesting functionals have expansions $F[f_a] = \sum \int (dx) F_{a_1,...,a_n}(x_1,...,x_n) f_{a_1}(x_1)....f_{a_n}(x_n)$ such that $F_n$ have regular branching at the ends, i.e. they behave as $L^\alpha$ where $L$ is the local equation of the end.

# 8 Bundles with operator structure algebras on open holomorphic manifolds.

In this section, we formalize the intuitive idea of "integrating out" a subspace of fields as it appears in physical literature. We will work with spaces of fields that have only regular singularities at the ends of the underlying manifold $M$. In coordinate space, singularities have fewer components, but the singularities are of irregular type.

We consider the functions that in local coordinates are given by convergent expansions near the end $L$

$$f_a(x) = \sum_{n \in \mathbb{N}} f_{a,g,n}(x_\parallel) L^{\alpha_{a,g}+n} + reg(x) \tag{8.37}$$

where $x_\parallel$ are coordinates on $L$, $f_{a,g,n}(x_\parallel)$ are regular at the generic point of $L$, $g \in \pi_1(M)$ denotes an element of the fundamental group of the manifold $M$. In general, we consider functions where the exponents $\alpha_{a,g}$ can depend on both the bundle index $a$ and on the fundamental group element $g$, i.e. they may be different on different sheets of analyticity.

We consider Wilsonian procedure for the functionals of the form

$$Z\{F\} = \int exp(-\int f(p) f(q) A(p,q) dp\ dq) F[f] Df \tag{8.38}$$

where $A(p,q)$ is holomorphic function of complex variables $p, q$ and $F[f]$ is a holomorphic functional. This is an expansion of the functionals we considered above. It is a generalization of the functionals considered by Wilson, which correspond to $A(p,q) = A(p)\delta(p,q)$. We suppressed the vector index $a$ of $f_a$.

We will assume that $A(p,q)$ have only regular singularities and the same is true about the coefficients of $F[f]$. Then, it is known that the coefficients of expansion of $Z\{F\}$ also have this property. The singularity divisor has infinitely many components $L_I$. The set of $L_I$ is closed with respect to the elimination operation, because the basic mechanism of singularity formation in these systems involves vanishing cycles.

The above discussion motivates consideration of manifolds of the type $\mathbb{C}^n - \cup L_I$ where $L_I$ are infinite collections of complex manifolds closed with respect to the elimination operators defined from the singularity set of $A(p,q)$.

We split the function as $f = g + h$ where $f, g$ are sub-bundles. We split the set of singularities $L_I$ into the sets $L'_I$ and $L''_I$. We also split the spectrum of singularities $\alpha_s$ into two sets $\alpha'_s$ and $\alpha''_s$. The bundles $g, h$ consist of holomorphic sections of bundles that can have regular singularity along $L_I$ with index $\alpha$ in one of the sets $A, B$.

**Def** LRG transformation that corresponds to a self similarity transform $\mathbb{C}^n - \cup_{I\in A\cup B} L_I \to \mathbb{C}^n - \cup_{I\in A} L_{I_s}$ and self similar splitting of the bundle $\oplus_{A\cup B} P_\alpha = (\oplus_{\alpha\in A} P_\alpha)\oplus (\oplus_{\alpha\in B} P_\alpha)$ is the transformation of the functional $E$ as in the formula 3.14.

# 9 Partial integration in QED and interaction representation.

Current formalism for computing higher order corrections to energy levels of bound states [17,52,53] is not LRG invariant. When we restore LRG invariance, it is necessary first to discuss the quantities we compute. The current formalism, as in cited papers, is focused on energy levels. However, energy levels are approximate notion due to the instability of these states. Since we are dealing with exact theory here, it is important to note that what is in principle computable is the lineshape curve of the state. It is this lineshape that should be LRG invariant. The energy levels are then the positions of the maxima associated with resonances. So in effect we are dealing with amplitude $< P|P' >$ where the final state, in the case of pure QED, contains also additional soft photons. This is an object of mixed space-time dimensionality, as usual in the treatments of Final State Radiation. This final state is expression of the following form $\{F_{\mu_1,...,\mu_n}(y, x_1, ..., x_n)[a, b, c]\}$ where $x_i$ are variables associated to soft photons and final stable particles( in the case of pure QED - only photons and electrons/positrons, as no other states have infinite lifetime).

We note the following implication of the present formalism. It is customary in perturbative calculations to use pure photon and electron states. This is what is necessary to construct Feynman diagrams. In LRG treatment, such notions are blurred. When one integrates out fields one is inevitably left with expressions of the form $\{F_{\mu_1,...,\mu_n}(y_1, ..., y_m, x_1, ..., x_n)[a, b, c]$ where $a, b, c$ are electron/position and photon creation operators, and $x_i, y_i$ are spacetime variables with corresponding tensor structures which we denote by $\alpha_i, \mu_j$ etc. The field operators depend on the underlying manifold of momenta which varies under the LRG transformation. Photons and other pure states become replaced by complexes of functionals.

We arrive at the following conclusion: LRG invariance is homotopy invariance under homotopy of a complex of function spaces

$$\begin{aligned} ... \to \Omega_{\alpha_1,...\alpha_p,\beta_1,...,\beta_q,\mu_1,...,\mu_r}(x_1, ..., x_p, y_1, ..., y_q, z_1, ..., z_r)[a, b, c] \to \\ \to \Omega_{\alpha_1,...\alpha_{p+1},\beta_1,...,\beta_{q+1},\mu_1,...,\mu_{r+1}}(x_1, ..., x_{p+1}, y_1, ..., y_{q+1}, z_1, ..., z_{r+1})[a, b, c] \to ... \end{aligned} \tag{9.39}$$

The arrows in this complex are operators of the form

$$K_{p,q,r}(x_1, ..., x_p, y_1, ..., y_q, z_1, ..., z_r; x'_1, ..., x'_{p+1}, y'_1, ..., y'_{q+1}, z'_1, ..., z'_{r+1})[a, \frac{\delta}{\delta a}, b, \frac{\delta}{\delta b}, c, \frac{\delta}{\delta c}] \tag{9.40}$$

where the exact form of these operators depends on the choice family of profinite manifolds. These operators involve integral kernels with regular singularities, and functional derivatives in the fields.

We would like to make the following finite dimensional analogy of the above developments. Consider a finite dimensional manifold $M$ and a set of forms $\omega^{(k)} \in \Omega^k(M)$. We consider equations that mix the forms $\omega^k$ as $\sum_\beta \hat{L}^\beta_\alpha \omega_\beta + \sum_{\sum \alpha_i = \alpha} \prod_i \omega_{\alpha_i} = 0$ and

we are interested only in such equations that are invariant under a class of homotopy transformations of the complex $\Omega^k(M)$. Above we deal with infinite dimensional analogy of this situation.

Renormalization group was considered in the context of field theories on Lorentz manifolds in [54, 55]. There are similarities of our approach and the approach in this literature. The relationship with operadic techniques was discussed in [9].

# 10 Forms vs tensor equations on function manifolds

When considering the manifold of functions $\mathbb{M} = \{A^a_\mu(x)\}$ there are several types of tensor objects that we can consider on it. The most natural one is antisymmetric $F_{\mu_1,...,\mu_n}(x_1, ..., x_n)[A]$ where there is antisymmetry in coordinates $x_i$ and indices $\mu_i$. This type of functionals corresponds to forms on $\mathbb{M}$. We are led to hypothesise that $F_{\mu_1,...,\mu_n}(x_1, ..., x_n)[A]$ and also slightly more general objects

$F_{\mu_{1,1},...,\mu_{1,s_1};...;\mu_{n,1},...,\mu_{n,s_n}}(x_1, ..., x_n)[A]$ realize every cohomology class of $\mathbb{M}$ even after we add infinity at its metric boundary. This is quite analogous to loop spaces and Chen iterated integrals( although at the time of writing even the analogies for maps from surfaces are not known).

However, such objects are not part of the discussion in current pheno literature. It is more expected to encounter functional tensors $F_{\mu_1,...,\mu_n}(x_1, ..., x_n)[A]$ that are symmetric in $x_i$. This would correspond to bosonic symmetry. We are led to hypothesise that such symmetric objects correspond to multi gluon wave functions. We are led to consider the following equations

$$\hat{L}(x_i, \frac{\partial}{\partial x_i})F_{\mu_1,...,\mu_n}(x_1, ..., x_n)[A] = \sum_m \hat{\hat{L}}(x_i, \frac{\partial}{\partial x_i}, y_j, \frac{\partial}{\partial y_j})[A, \frac{\delta}{\delta A}]F(y_1, ..., y_m)[A], \tag{10.41}$$

where $\hat{L}$ are ordinary differential operators and $\hat{\hat{L}}$ are functional differential operators. There are also non linear versions that contain non linear terms of the form

$$NL = \int_{z_1,...,z_k} \prod F_s(x, y, z)[A] \tag{10.42}$$

where we symbolically wrote $F_s(x, y, z)[A]$ for terms of the form $F(x_{i_1}, ..., y_{j_1}, ..., z_{k_1}, ...)[A]$ in various possible combinations.

We would like to make the following important observation. All of these equations have formal perturbative solutions that superficially look like Feynman intgrals. They are easily encompassed by configuration space integrals, and modules over these algebras, encountered in topology [26, 56–59]. This means that we need some symmetry principle to restrict our equations.

# 11 Categorical formulation of renormalization group

In this section we discuss categorical formulation of renormalization group. Considering a QFT as a functor on a category of manifolds has a long history( for recent

developments see [60–62]). This approach works well for topological QFTs, while QCD has explicit dependence on metric tensor. As we noted above, local renormalization leads to arbitrary metric structures. Therefore, in this section we address the problem of considering QFT on the category $m\bar{a}n(4)$ of topological 4-manifolds. Moreover, we would like to study the derivative $\delta/\delta g_{\mu\nu}(x)Z[g;A,\psi]$ of the QFT as a functional of metric. We would like to make this notion precise.

We start with the observation that effective actions $S[g;f]$ for a given collection of fields are not necessarily restricted to the category of smooth manifolds. The manifold structure is allowed to have mild singularities, as long as the integrations in nonlocal action make sense. For example, action is defined on the subcategory of manifolds with Holder structures. Singularities in this subcategory are allowed to have the form $L(x)^{\alpha}$ for some complex $\alpha$. This is a strict subcategory of $m\bar{a}n(4)$.

When we discuss the variational derivative $\delta/\delta g_{\mu\nu}(x)$ it is important to discuss which function space $g_{\mu\nu}$ is allowed to vary over. There are many natural classes of function spaces [63]. We arrive at the necessity to consider explicit variables that describe the function space. The proper language for this description is descriptive set theory. Such description is detailed in [64]. We can consider the function space information to be collected into a symbol $\mathbb{L}$. roughly speaking, this symbol contains information which limits of the functions on manifolds exist. For example, $\mathbb{L}$ contains as a subset the constructive set of Holder functions with singularities that are localized on a subcomplex. We arrive at the following conclusion. QFT is a map from the space of definable metrics to the target set

$$Z : Riem^{\mathbb{L}}(M) \to S \tag{11.43}$$

where $S$ can in simple situations to be restricted to reals, but in most general situations is a larger set of model theoretic origin [65–68].

We now show how this statement intertwines with the statement on manifold dependence. Surgery on topological manifolds can be realized as flows of metric structures that develop non local singularities. Therefore, such surgeries naturally embed in the space $Riem^{\mathbb{L}}(R^d)$, with appropriate choice of $\mathbb{L}$.

The above discussion shows that QFT can be considered as a map

$$Z : Diff^{\mathbb{L}}(M) * Riem^{\mathbb{L}}(M) * F^{\mathbb{L}}(M) \to S \tag{11.44}$$

for some choice of $\mathbb{L}$. The question arises on the exact structure of the domain. Extension of the domain may necessitate corresponding extension of the range to accommodate new ways to take limits in $\mathbb{L}$. We note that in a natural sense the domain contains the category of discrete groups $\bar{\pi_1}$ because each discrete group is realizable as fundamental group of a 4-dimensional manifold. We also note that $Z$ admits expansion to Gromov-Hausdorff space(GH for short) because GH is a limit set of $Riem^{\mathbb{L}}(R^4)$.

We now address the question how the above picture looks like in the most mundane case of QED. However, it is necessary to extend the definition of QED to include: 1) Lorenzian manifolds with arbitrary cylindrical ends $\bar{M}$ 2) formulate the theory on arbitrary curved background.

We note that the notion of momentum in QED is tightly related with the end structure of the manifold. The fact that there is a 3-momentum that belongs to a 3d hyperbolic manifold is a consequence of the fact that light cone in Minkowski space has two cylindrical ends, and the 3d momentum is a coordinate on these ends.

With the extensions above, we obtain $Z_{QED}[f, e, A, \psi]$, where we take tetrad field $e$ instead of metric tensor $g$ to allow formulation for spinor fields. This functional acts as

$$Z_{QED} : Diff^{\mathbb{L}}(\bar{M}) * Riem^{\mathbb{L}}(\bar{M}) * A^{\mathbb{L}}(\bar{M}) * P^{\mathbb{L}}(\bar{M}) \to S \tag{11.45}$$

where $A(\bar{M})$ and $P(\bar{M})$ are spaces of fields consistent with asymptotic cylindrical structure near the ends. Local renormalization group corresponds to geometric flow with surgeries that are consistent with the end structure.

**Rk:** In the case of 4-manifolds, there was a family of 4-manifolds exhibited in [69], for which the ends belong to a discrete family of 4-manifolds and corresponding manifold structures can be reparametrized by a finite tower of logarithmic functions. This situation matches very well with the usual perturbative calculations in single scale QED [70], and it would like to clarify how this family of manifolds arises in QED phenomena. In particular, what is the role of regularized mass $m_e(log(\mu))$, considered as curvature of the hyperbolic 3-manifolds in the construction of [69].

As final remarks in this section, we would like to discuss the relationship of our formalism with the recent works on topology of spaces of diffeomorphisms [26, 71–74]. As we argued above, QFT is a theory of functional vector bundles on spaces that are locally modelled on submanifolds of spaces like $Diff_c(R^n)$. For example, this definition encompasses submanifolds of $Diff_\partial(D^n)$ of diffeomorphisms of the disk, fixing the boundary, i.e. spaces considered in [74]. QFT, in particular QED, should be definable on these function manifolds as a theory of spinor bundles, generalizing the framework of [27], based on bundles of rigged Hilbert spaces. But actually QFT as we define it is more flexible. It is for example defined on $\mathbb{M} \subset Diff_\partial(D^n)$ of infinite codimension. For geometirc considerations about the type of manifolds that can arise in this context see [75].

However, we would like to restrict our attention to topological situations. The reason for this is that in this case the tangent bundle of the functional manifolds is in appropriate sense quantized, for example, in the case of $Diff_\partial(D^n)$ it has the structure of a spectrum [26, 74]. There are similar examples such as $BTop(d)$ [26]. We therefore arrive at the following problems:

**Q1:** Define a natural class of submanifolds of function spaces for which the tangent space is a spectrum.

**Q2:** Define the notion of functional QFT, on function manifolds of the above type, which are insensitive to metric structure of the function manifold.

Problem 2 naturally extends the notion of QFT into the theory of bundles on a general class of function manifolds.

We note that QED has the above property 2. It would be interesting to study the problem of Gribov copies through this lens [76].

Problem 1 is partially motivated by the work [59], where Kontsevich-type inetrgals were used to construct invariants of diffeomorphism group of $R^4$. Problem 1 essentially asks for the type of function manifolds for which the tangent bundle admits definition of similar constructive set of invariants. We note that this question is very well motivated by the intagral formulas widely used in the context of QCD factorization [77, 78]. We hypothesise that various factroization formulas in QCD can be unified using the above formulation using spectra, and their corresponding integral invariants.

# 12 Functional homotopy invariance and the Dirac spectrum.

We would like to discuss renormalization of Dirac spectrum. This system is of immediate experimental interest [23, 24, 79, 80] and enormous amount of theory work went into understanding it [81–86]. We start with the observation that spectrum is not characterized by single numbers $E_{n,l,m,j}$ as is customary in the literature [23]. In fact, this spectrum is a function $\rho(E)$ of the energy of the emitted photon. This function has approximately Lorenz shape, from which $E_{n,l,m,j}$ is usually inferred [23]. It is the function $\rho(E)$ that has physical significance. We argue in this section that this function is a homotopy invariant, under functional homotopy applied to the space of fields $\mathbb{M}$.

We make a few remarks on the functional manifold $\mathbb{M}$. This manifold is not locally compact, and is not compactifieable. It can be contrasted to the space of Lipschitz functions between compact differentiable manifolds. This space is compact [87]. In contrast, the space of fields has ends, that generalize the notion of ends in finite dimensions [88, 89]. In contrast to two dimensional situations [90], the pseudo compactifications( additions of geometric ends) lead to spaces such that open covers in general only admit subcovers of cardinality continuum( in the classical situations, countable covers do exist). There are several ways to define the notion of homotopy invariance for such generalized manifold. One way is to cover the manifold with simplices( continuum of these, usually). The simplices can be made Hilbertian( or any $l_p$) with countable subdivisions. Then homotopy is defined simlicially, by subdividing, and contracting some faces. This is a functional analogy of simple homotopy theory [91]. Another way is more geometric, and goes through parametrization of manifold and collapsing cells in the course of a flow. We will adopt the second approach below.

At leading order, the Dirac spectrum emerges from the poles of the solution of the Schwinger equation for the Green's function [92]. This approach allows to factorize the Green's function as $G_n(\vec{p}, \vec{q})$ so that the corresponding residue - the wave function $\psi_n(\vec{p})$ - behaves as $\psi(\vec{p}) \sim P_n(\vec{p}(\vec{p}^2 + a_n^2(\alpha(\mu), m_e(\mu)))^{\gamma_n(\alpha(\mu))}$ where $a_n$ is the Dirac energy and $\gamma_n(\alpha)$ is the appropriate power law. Our immediate goal is to generalize this behavior to be more in line with our analysis. First, at higher orders it is necessary to consider higher component wave functions $\psi_n(t; \vec{x}_1, ..., \vec{x}_n)$ or their multitime versions $\psi_n(x_1, ..., x_n)$. According to our master evolution equations, these wave functions should be considered as Taylor coefficients of the wave functional $F$. This functional results from solutions of the evolution equation, consistent with factorization of the functional near the ends of the manifold of fields $\mathbb{M}$. The Taylor coefficients have an algebra of singularities, that is basically determined by the structure of singularities of the evolution kernel( there is a finite dimensional freedom). We therefore arrive at the interpretation that Dirac spectrum arises dynamically as a solution of functional evolution equations. The simple Dirac expression corresponds to the lowest component of the wave functional expansion.

Now we interpret renormalization group transformations as homotopy invariance. For this, we interpret the simplices in the simplicial subdivision of $\mathbb{M}$ as being realized by holomorphic functions of complex coordinates, with moving singularities. Then, we can explicitly write the homotopy transformation of $\mathbb{M}$, simplex wise, as application of an integral operator $S(x, y)$. The crucial recuirement is that this integral kernel $S(x, y)$ should have singularities consistent with the singularity algebra of the wave functional

$F$, and the singularity types present in $S$ must be consistent with singularity types in $F$. In this case, we interpret homotopy transformation as application of the singular integral kernel $S$. Such application results in the transformation of the components of the singularities of $F$. This is precisely renormalization group flow of the functional $F$. It must be accompanied by a flow in the factorization coefficients. According to our discussion in previous sections, these coefficients are sections of - usually infinite dimensional - bundles, that have structure group consistent with group $C^*$-algebra of the fundamental group of the complement of singularities.

# 13 Local renormalization group and first principle the proton mass calculation.

In this section we discuss LRG invariance of field equations for bound states. We build LRG invariance into the QFT evolution equations. The problem we face here is that full equations that incorporate both $g-2$ calculation [93] and the multiloop corrections to Dirac spectrum are not known at the moment. There exists large literature devoted to multi loop calculations for Lamb shift, e.g. [17, 52, 53]. At the order of $\alpha log(Z\alpha)^{-1}$ there is agreement with experiment [23, 94]. But the problem is that these equations are ad hoc. For example, the formulas in [95, 96] explicitly refer to bare wave function of the electron, while in first principle computation, it should be possible to obtain the whole tower of spinor wave functions from single dynamical equation. The approach in the above papers relies on truncation of Bethe Salpeter, see the discussion in [97]. This reference relies on the introduction of Coulomb propagators. While numerically satisfying, it is conceptually unsatisfactory. We propose to consider the following evolution equation

$$i\frac{\delta}{\delta t^{LRG}_{\mu\nu}(x)}F = T^{LRG}_{\mu\nu}(x)[a, \frac{\delta}{\delta a}]F \tag{13.46}$$

where $t^{LRG}_{\mu\nu}(x)$ is a functional time variable, and $T^{LRG}_{\mu\nu}(x)$ is a covariant generalization of RG evolution hamiltonian. This equation generalizes the QCD evolution equations DGLAP and BFKL.

The above equation is related to factorization algebras [8, 98–100]. The functionals $F[a] = F_{\alpha_1,...,\alpha_n}(x_1, ..., x_n)[a]$ are holomorphic except for singularities that have parametric equations

$L_I(x_1, ..., x_n; y_1, ..., y_m)[t]$ for the m-th component $F(x_1, ..., x_n; y_1, ..., y_m)$. The singularities $L_I$ move( evolve) with $t(x)$. This is the local renormalization group flow. The functionals are (multidimensional) transseries( in this way we relate to the earlier work [101, 102] on 1-dimensional transseries). Locally they can have singularities of logarithmic type $s^a log^n(s)$, where $s = L_I$. At the divisor at infinity, they can have exponential singularities $exp(s^a)$, or generalized expoential singularities $exp(poly(s, exp(s_i^a))$. The full classification of singularities depends on the quantization procedure, and the type of the singular integral kernels we put into $T_{\mu\nu}(x)[a, \frac{\delta}{\delta a}]$. The types of terms allowed in this evolution kernel is related to the structure of the manifold of fields $a$. This structure is tigtly related to the the choice $\hat{A}(x)[a, \frac{\delta}{\delta a}$ and also the the structure of the tangent bundle $\mathbb{T}_a\mathbb{M}$ to the function manifold $\mathbb{M} = \{a\}$. These are two independent choices. The second has to do with a fenctional analogy of quantization of the tangent bundle of Kahler manifolds [103].

The above equation generalizes the Light Cone QCD formalism of Brodsky et. al. [4–6]. With suitable choice of the hamiltonian, with restriction to single time( i.e. choosing $t(x) = t = const$) we ontain the LC QCD equation $H_{LC}F = EF$. One more important distinction from the LCQCD lies in the treatment of functional tensors $F_{\alpha_1,...,\alpha_n}(x_1,...,x_n)[a]$. We discussed the necessity of such tensors in the previous section. Phenomenologically, this leads to the imortant distinction of the types of variabales in Taylor coefficients $F(x_1,...,x_n;y_1,...,y_m)$. In variables $y_i$ the functional is symmetric, while in variables $x_i$ it can be bot symmetric and antisymmetric. The antisymmetric case corresponds to forms on the functional manifold. This allows for alternative treatment of Fermion symmetry: if the variable $a$ is the occupation number field of the fermion field, then the function $a$ is still an element of the usual function space of sections. The Fermi anti-commutation rules are recovered by requiring anticommutation on the level of variables $x_i$ in the functional. Therefore, we recover fermions if we consider forms on the functional manifold of fields. We recover bosons by upgrading the gauge fields to the non-local algebra of operators $\hat{D}(x,y)[A]$ acting pointwise on the manifold of fermion variables.

Our equation generalizes both DGLAP and BFKL equations. These equations are good for inclusive observables of the form $\int (dp_i) < F_1 *_{Q^2} F_2 | \hat{\hat{O}}[a, \frac{\delta}{\delta a}] | \prod G_i(p_i) >$ where the integration is performed over most of the momenta $p_i$ of the final state paeticles, described by the functionals $G_i[a]$. After this integration, single renormalization scale is selected. The kernels corrspond to resummation of forward partons. We recover the results of this resummation in the following way. The resulting function $f(x,Q^2)$ - the solution of BFKL or DGLAP equation with power law initial conditions [104–106] - is a resurgent function of both $x$ and $Q^2$. We recover the resurgent structure of this function from our evolution equation. It corresponds to taking the iterated convolutuon of functionals, and then combining the evolution of the leading coefficients near the singularities $L_I(x_1,...,x_n;y_1,...,y_m)[t]$. In more detail, the functionals $F_i$ or $G_i$ factorize( as transseries) as $F = (L_I)^a * F_I(y_{\|}) + regular$, in the case of regular singularities( this is all that is needed for DGLAP and linear BFKL), where $y_{\|}$ are the coordinates on the singularity manifold. Then our equation predicts the evolution of both singularity locus $L_I$ and the factorization coeffcient $F_I$. Therefore, we interpret the singularities in the compexified space of $x, Q^2$ as arising from higher dimensional singularities of the Taylor coefficients of functional $F$, after we perform suitable convolutions to obtain the inclusive observable. In a sense, the singularities in $x, Q^2$ space are invariants of the local renormalization group motion in the full functional space.

The relationship with the traditional DGLAP and BFKL evolution equations is the following. These latter equatiions correspond to emission of partons in a restricted kinematics, which then get resummed by the proton pdfs. There is single evolution scale in the process - the energy transfer scale. In our formalism, we consider the whole tower of partonic states, i.e. the collection of functions of partonic momenta $\vec{p_i}$. These functions are locally expressible as Puiseux series with singularities on a divisor in the space of partonic momenta. Traditional RG evolution allows to fix the exponents in Bjorken $x$ of regular solutions in terms of the coupling at a scale. The equation above concerns genralization of this to multiparameter regular dependence. For each of the components $L_i(p_1,...,p_n)$ of the singularity divisor, there is its own leading branching exponent. The equation above fixes these exponents in a self consistent manner.

Perturbatively, we expact the local time to be expanded as

$$t_{\mu\nu}^{LRG}(x) = \sum \alpha^n t_{n,\mu\nu}^{LRG}(x) \tag{13.47}$$

where $t_{n,\mu\nu}^{LRG}(x)$ can be found as functional eigenvalues, as we detail in an example below. Here we draw the analogy with DGLAP and BFKL further. In the latter case, there is single renormalization scale. We determine the local scale parameter completely from the problem. Due to self consistency of the first order equation, we expect that we can choose a global field of scales $t^{LRG}(x)$, in terms of a single evolution parameter. This is analogous the choice if single time in multitime evolution equations $\partial/partialt_i f(t_1, ..., t_n) = H_i(t_1, ..., t_n)f$: if the equaitons are consistent, we can choose $t_1 = ... = t_n = t$. This overall scale translates into the RG scale in the traditional DGLAP equations.

We also provide detail on the mathematical meaning of $F$. While it is generally known that $F$ is a Fock state vector, we argue that it carries more structure. $F$ is a section of a bundle defined on the function space of sections of occupation numbers $a(\vec{p})$, where $a$ runs over the complete set of fields in the theory(in the case of QED, it is electrons/positions and photons). In components, in local chart

$$F = \sum_n F_{A_1,...,A_n}(x_1, ..., x_n)[a] \tag{13.48}$$

where $A_i$ are tensor indices accociated to the variable $x_i$.

In order to study the properties of solutions of the equation 13.46, we introduce a toy model. This toy model has the form

$$\hat{\hat{\mathbb{L}}}(x)[\frac{\delta}{\delta f}]F = 0 \tag{13.49}$$

This equation is stationary version of the full evolution equation 13.46. In this equation, the variational operator has constant coefficients. This is functional analogy of the finite dimensional hyperbolic operatrs with constant coefficients [107]. We expect the presence of functional analogies of Petrovsky lacunas.

In order to further constrain the system, we are going to assume that the coefficients of this equation are resurgent [108–110], with regular local singularities. The resurgence condition for a function $f_g(z)$ of a complex variable $z$ means the following package of analytic data:

1) there is at most countable set of points $z_n$ where the function $f(z)$ can have singularity. The points may be dense in projection, but after the lift to the universal cover of their complement, from $z_n$ only finitely many $z_m$ are reachable in finite distance on the Riemann surface of $f(z)$. The Riemann surface of $f(z)$ can have infinite type.

2) near each lift of $z_n$, the function has a local Puiseus expansion $f_g(z) = \sum f_n(z - z_n)^{\alpha_{g,n}} reg_g(z) + reg_g(z)$, where the exponent $\alpha_{g,n}$ may depend on the lift. In principle, such expansions can have logarithms, but we assume a version of dimensional regularization, which lifts the logarithmic terms into a log free Puiseux form of the above type.

**Th:** If the set $x$ is the complex plane $\mathbb{C}$ and the coefficients of $\hat{\hat{L}}_x[\delta]$ are resurgent with regular singularities, then there are isolated solutions of the equation 13.49 that are also resurgent with regular singularties.

In order to prove this theorem, we make the following observation. The solutions 13.49 under genericity assumption can be sought in the exponential form

$$F[f] = exp(\int k_x f_x) \tag{13.50}$$

for a function $k_x$ that satisfies the following functional equation

$$L(x)[k] = 0 \tag{13.51}$$

This is an example of equation of polynomial type in functional space. We are considering perturbative solutions of this equation. Suppose that locally this equation can be written in the form

$$L_0(x) + \int_\zeta dy L_1(x,y) f(y) + o(f) = 0 \tag{13.52}$$

In this equation, $L_0(x), L_1(x,y)$ are resurgent functions. The theorem follows from the fact that solutions to integral equations with resurgent kernels are also resurgent.

The physical meaning of this toy example is clarified with identification of the field $k_x$ with the energy scale field $T_{\mu\nu}(x)$. The energy momentum tensor is obtained as functional eigenvalue of the functional differential equation.

We constructed an isolated solution with the assumption 13.52. This is a non trivial assumption because one can construct examples where solutions have condensation points. The physical meaning of such solutions merits investigation.

There is certain interplay of our formalism with the widely discussed idea of effective spacetime [111,112]. We argue that this phenomenon is present already in very simple quantum systems, such as hydrogen atom, at multiple loops. In fact, it is already observed in experiments [113], as the spectral shape function of the photons. As a result of the above theorem, the LRG scale field is a self consistent solution of QED evolution equations with local renormalization scales. This effect is more pronounced in strongly coupled systems. Our formalism provides multiscale extension of the treatment of energy momentum tensor in [114,115]. The result is a resurgent model for the function $< F|\hat{T}_{\mu\nu}(x)|I >$, evaluated on the resurgent functionals for initial and final states of the process. We note that in canonical perturbation theory, is multidimensional version of Stokes type resurgence, with no worse than exponential singularities. It is work in progress by the author to clarify possible non perturbative effects and the type of singularities that may be encountered in non perturbative quantization.

# 14 Conclusion

In this paper we discussed conceptual framework for bringing the ideas of Local Renormalization Group into computations of structure of bound states in QFT. We suggest that spectral characteristics are homotopy invariants of functional complexes. We generalize the notion of homotopy from finite dimensional to functional complexes. Out theory provides a notion of homotopy suitable for study of functional forms on function manifolds. Our study suggests ample connections with the theory of Banach homological algebras. We discussed the interplay of our theory with the theory of resurgent functions with regular singularities. In coordinate space formulations, these are spaces of holomorphic functions functions with irregular singularities on non movable divisors, with fixed Stokes data. We hope that our discussion will foster a discussion of these mathematical ideas.

# References


[1] Joseph Polchinski. Renormalization and effective lagrangians. Nuclear Physics B, 231(2):269–295, 1984.

[2] Christof Wetterich. Exact evolution equation for the effective potential. Physics Letters B, 301(1):90–94, 1993.

[3] H Osborn. Weyl consistency conditions and a local renormalisation group equation for general renormalisable field theories. Nuclear Physics B, 363(2-3):486–526, 1991.

[4] Alexandre Deur, Stanley J Brodsky, and Guy F De Téramond. The qcd running coupling. Progress in Particle and Nuclear Physics, 90:1–74, 2016.

[5] Alexandre Deur, Stanley J Brodsky, and Craig D Roberts. Qcd running couplings and effective charges. Progress in Particle and Nuclear Physics, 134:104081, 2024.

[6] Stanley J Brodsky, Hans-Christian Pauli, and Stephen S Pinsky. Quantum chromodynamics and other field theories on the light cone. Physics Reports, 301(4-6):299–486, 1998.

[7] Kevin Costello. Renormalization and effective field theory. Number 170. American Mathematical Soc., 2011.

[8] Kevin Costello and Owen Gwilliam. Factorization algebras in quantum field theory, volume 2. Cambridge University Press, 2021.

[9] Jean-Louis Loday and Nikolay M Nikolov. Operadic construction of the renormalization group. In Lie Theory and Its Applications in Physics: IX International Workshop, pages 191–211. Springer, 2013.

[10] Christian Brouder, Alessandra Frabetti, and Frédéric Menous. Combinatorial hopf algebras from renormalization. Journal of Algebraic Combinatorics, 32(4):557–578, 2010.

[11] Mikael Rørdam, Erling Størmer, and M Rørdam. Classification of nuclear, simple C*-algebras. Springer, 2002.

[12] Benoit Bossard. Coanalytic families of norms on a separable banach space. Illinois Journal of Mathematics, 40(2):162–181, 1996.

[13] Hui-Hsiung Kuo. Gaussian measures in banach spaces. Gaussian measures in banach spaces, pages 1–109, 2006.

[14] Robert M Solovay. A model of set-theory in which every set of reals is lebesgue measurable. Annals of Mathematics, 92(1):1–56, 1970.

[15] Saharon Shelah. Can you take solovay's inaccessible away? Israel Journal of mathematics, 48:1–47, 1984.

[16] Edward Witten. Topological quantum field theory. Communications in Mathematical Physics, 117(3):353–386, 1988.

[17] Savely G Karshenboim, Akira Ozawa, Valery A Shelyuto, E Yu Korzinin, Robert Szafron, and Vladimir G Ivanov. The complete $\alpha 8$ m contributions to the 1 s lamb shift in hydrogen. Physics of Particles and Nuclei, 53(4):773–786, 2022.

[18] Savely G Karshenboim, Akira Ozawa, and Vladimir G Ivanov. Higher-order logarithmic corrections and the two-loop self-energy of a 1 s electron in hydrogen. Physical Review A, 100(3):032515, 2019.

[19] Gregory S Adkins and Ulrich D Jentschura. Irreducible three-loop vacuum-polarization correction in muonic bound systems. Physical Review D, 111(5):056016, 2025.

[20] Michael I Eides and Vladimir A Yerokhin. Energy levels of multiscale bound states from qed energy-momentum trace. Physical Review D, 113(5):056012, 2026.

[21] Jakub Hevler, Andrzej Maroń, and Krzysztof Pachucki. Relativistic nuclear recoil effects in hyperfine splitting of hydrogenic systems. Physical Review A, 113(4):042813, 2026.

[22] Vojtěch Patkóš, Vladimir A Yerokhin, and Krzysztof Pachucki. Two-body p-state energies at order $\alpha$ 6. Physical Review A, 109(2):022819, 2024.

[23] Alexey Grinin, Arthur Matveev, Dylan C Yost, Lothar Maisenbacher, Vitaly Wirthl, Randolf Pohl, Theodor W Hänsch, and Thomas Udem. Two-photon frequency comb spectroscopy of atomic hydrogen. Science, 370(6520):1061–1066, 2020.

[24] Lothar Maisenbacher, Vitaly Wirthl, Arthur Matveev, Alexey Grinin, Randolf Pohl, Theodor W Hänsch, and Thomas Udem. Sub-part-per-trillion test of the standard model with atomic hydrogen. Nature, pages 1–7, 2026.

[25] Stanislav Srednyak. Flat bundles on function manifolds and evolution equations in quantum field theories. Foundations, 6(2):19, 2026.

[26] Manuel Krannich and Oscar Randal-Williams. Diffeomorphisms of discs and the second weiss derivative of btop (-). arXiv preprint arXiv:2109.03500, 2021.

[27] Peter Kristel and Konrad Waldorf. Smooth fock bundles, and spinor bundles on loop space. Journal of Differential Geometry, 128(1):193–255, 2024.

[28] Mauro Spera and Tilmann Wurzbacher. Twistor spaces and spinors over loop spaces. Mathematische Annalen, 338(4):801–843, 2007.

[29] Christian Bonatti, Lorenzo J Díaz, and Marcelo Viana. Dynamics beyond uniform hyperbolicity: A global geometric and probabilistic perspective. Springer, 2005.

[30] Marco Martens, Liviana Palmisano, and Zhuang Tao. Newhouse laminations of polynomials on $\mathbb{C}^2$. International Journal of Mathematics, 31(11):2050091, 2020.

[31] Manuel Krannich and Alexander Kupers. Infinity-operadic foundations for embedding calculus. Journal of Topology, 19(2):e70071, 2026.

[32] Michael H Freedman and Laurence R Taylor. A universal smoothing of four-space. Journal of Differential Geometry, 24(1):69–78, 1986.

[33] James E Hanson. A metric set theory with a universal set. The Journal of Symbolic Logic, pages 1–34, 2023.

[34] Kathryn Mann. The structure of homeomorphism and diffeomorphism groups. Notices of the American Mathematical Society, 68(4), 2021.

[35] Kathryn Mann and Christian Rosendal. Large-scale geometry of homeomorphism groups. Ergodic Theory and Dynamical Systems, 38(7):2748–2779, 2018.

[36] Maxim Kontsevich and Yan Soibelman. Holomorphic floer theory i: exponential integrals in finite and infinite dimensions. arXiv preprint arXiv:2402.07343, 2024.

[37] Sebastian Mizera and Simon Telen. Landau discriminants. JHEP, 08:200, 2022.

[38] Franc Forstnerič. Stein manifolds and holomorphic mappings. Springer, 2011.

[39] Danny Calegari. Foliations and the geometry of 3-manifolds. Clarendon press, 2007.

[40] Alberto Candel and Lawrence Conlon. Foliations II, volume 2. American Mathematical Soc., 2000.

[41] Piotr Przytycki, Gabriela Schmithüsen, and Ferrán Valdez. Veech groups of loch ness monsters. In Annales de l'Institut Fourier, volume 61, pages 673–687, 2011.

[42] Yasmina Atarihuana, Juan García, Rubén A Hidalgo, Saúl Quispe, and Camilo Ramírez Maluendas. Dessins d'enfants and some holomorphic structures on the loch ness monster. The Quarterly Journal of Mathematics, 73(1):349–369, 2022.

[43] John A Arredondo and Camilo Ramirez Maluendas. On infinitely generated fuchsian groups of the loch ness monster, the cantor tree and the blooming cantor tree. Complex Manifolds, 7(1):73–92, 2020.

[44] Daniel Allcock. Most big mapping class groups fail the tits alternative. Algebraic & Geometric Topology, 21(7):3675–3688, 2021.

[45] Kathryn Mann and Kasra Rafi. Large-scale geometry of big mapping class groups. Geometry & Topology, 27(6):2237–2296, 2023.

[46] Javier Aramayona, Rodrigo De Pool, Rachel Skipper, Jing Tao, Nicholas G Vlamis, and Xiaolei Wu. Non-planar ends are continuously unforgettable. International Mathematics Research Notices, 2025(5):rnaf026, 2025.

[47] Grigory Mikhalkin. Decomposition into pairs-of-pants for complex algebraic hypersurfaces. Topology, 43(5):1035–1065, 2004.

[48] Mudumbai S Narasimhan and Conjeeveram S Seshadri. Stable and unitary vector bundles on a compact riemann surface. Annals of Mathematics, 82(3):540–567, 1965.

[49] Simon Albrecht and Christopher J Bishop. Speiser class julia sets with dimension near one. Journal d'Analyse Mathématique, 141(1):49–98, 2020.

[50] Rostislav I Grigorchuk. Degrees of growth of finitely generated groups, and the theory of invariant means. Mathematics of the USSR-Izvestiya, 25(2):259, 1985.

[51] Charles Favre and Mattias Jonsson. Valuations and multiplier ideals. Journal of the American Mathematical Society, 18(3):655–684, 2005.

[52] Krzysztof Pachucki, Vojtěch Patkóš, and Vladimir A Yerokhin. Three-photon-exchange nuclear structure correction in hydrogenic systems. Physical Review A, 97(6):062511, 2018.

[53] B Ohayon and Ulrich D Jentschura. Reexamination of vacuum-polarization corrections to the self-energy in muonic bound systems. Physical Review A, 110(3):032820, 2024.

[54] Edoardo D'Angelo and Kasia Rejzner. A lorentzian renormalization group equation for gauge theories. In Annales Henri Poincaré, volume 26, pages 4411–4459. Springer, 2025.

[55] Klaus Fredenhagen and Katarzyna Rejzner. Batalin-vilkovisky formalism in perturbative algebraic quantum field theory. Communications in Mathematical Physics, 317(3):697–725, 2013.

[56] Julien Ducoulombier and Victor Turchin. Delooping the functor calculus tower. Proceedings of the London Mathematical Society, 124(6):772–853, 2022.

[57] Simon Brun and Thomas Willwacher. Graph homology computations. arXiv preprint arXiv:2307.12668, 2023.

[58] Michael S Weiss. Rational pontryagin classes of euclidean fiber bundles. Geometry & Topology, 25(7):3351–3424, 2022.

[59] Tadayuki Watanabe. On kontsevich's characteristic classes for higher dimensional sphere bundles i: the simplest class. Mathematische Zeitschrift, 262(3):683–712, 2009.

[60] Daniel Grady and Dmitri Pavlov. Higher categories of bordisms with geometric structures. arXiv preprint arXiv:2605.03453, 2026.

[61] Urs Schreiber. Aqft from n-functorial qft. Communications in Mathematical Physics, 291(2):357–401, 2009.

[62] Daniel S Freed, Gregory W Moore, and Constantin Teleman. Topological symmetry in quantum field theory. Quantum Topology, 15(3):779–869, 2024.

[63] Hans Triebel. Theory of function spaces III. Springer, 2006.

[64] Bruno de Mendonca Braga, Willian Hans Goes Correa, and Valentin Ferenczi. Descriptive set theory of separable frechet spaces. arXiv preprint arXiv:2508.09368, 2025.

[65] Alessandro Berarducci and Vincenzo Mantova. Transseries as germs of surreal functions. Transactions of the American Mathematical Society, 371(5):3549–3592, 2019.

[66] Matthias Aschenbrenner, Joris van der Hoeven, and Lou Van den Dries. Asymptotic differential algebra and model theory of transseries:(ams-195). 2017.

[67] Ehud Hrushovski and Masanori Itai. On model complete differential fields. Transactions of the American Mathematical Society, 355(11):4267–4296, 2003.

[68] David Aspero and Konstantinos Tsaprounis. Long reals. Journal of Logic and Analysis, 10, 2018.

[69] Michael Freedman and Michael Starbird. The geometry of the bing involution. arXiv preprint arXiv:2209.07597, 2022.

[70] Riccardo Aliberti, Tatsumi Aoyama, Elisa Balzani, Adnan Bashir, Genessa Benton, Johan Bijnens, Volodymyr Biloshytskyi, Thomas Blum, Diogo Boito, Mattia Bruno, et al. The anomalous magnetic moment of the muon in the standard model: an update. Physics Reports, 1143:1–158, 2025.

[71] Ben Knudsen and Alexander Kupers. Embedding calculus and smooth structures. Geometry and Topology, 28(1):353–392, 2024.

[72] Alexander Kupers and Oscar Randal-Williams. On diffeomorphisms of even-dimensional discs. Journal of the American Mathematical Society, 38(1):63–178, 2025.

[73] Alexander Kupers. Three applications of delooping to h-principles. Geometriae Dedicata, 202(1):103–151, 2019.

[74] Manuel Krannich and Alexander Kupers. The disc-structure space. In Forum of Mathematics, Pi, volume 12, page e26. Cambridge University Press, 2024.

[75] Andrés Navas. Groups of circle diffeomorphisms. University of Chicago Press, 2011.

[76] Nele Vandersickel and Daniel Zwanziger. The gribov problem and qcd dynamics. Physics reports, 520(4):175–251, 2012.

[77] Andre H Hoang, Vicent Mateu, Matthew D Schwartz, and Iain W Stewart. Precision e+ e- hemisphere masses in the dijet region with power corrections. Journal of High Energy Physics, 2025(9):1–55, 2025.

[78] Daniele Bertolini, Daniel Kolodrubetz, Duff Neill, Piotr Pietrulewicz, Iain W Stewart, Frank J Tackmann, and Wouter J Waalewijn. Soft functions for generic jet algorithms and observables at hadron colliders. Journal of High Energy Physics, 2017(7):1–47, 2017.

[79] Omer Amit, Derya Taray, Vitaly Wirthl, Vincent Weis, Mustafa Waqar Syed, Akira Ozawa, Johannes Weitenberg, Savely G Karshenboim, JTM Walraven, Lothar Maisenbacher, et al. Towards trapping of hydrogen atoms for computable optical clock applications. Physical Review A, 112(3):033101, 2025.

[80] Pedro Amaro, Andrzej Adamczak, Marwan Abdou Ahmed, L Affolter, Fernando Domingues Amaro, Patricia Carvalho, T-L Chen, Luis MP Fernandes, M Ferro, Damian Goeldi, et al. Laser excitation of the 1s-hyperfine transition in muonic hydrogen. SciPost Physics, 13(2):020, 2022.

[81] S. G. Karshenboim, A. Ozawa, V. A. Shelyuto, E. Yu. Korzinin, R. Szafron, and V. G. Ivanov. The Complete $\alpha^8$m Contributions to the 1s Lamb Shift in Hydrogen. Phys. Part. Nucl., 53(4):773–786, 2022.

[82] Savely G. Karshenboim, Akira Ozawa, Valery A. Shelyuto, Robert Szafron, and Vladimir G. Ivanov. The Lamb shift of the $1s$ state in hydrogen: Two-loop and three-loop contributions. Phys. Lett. B, 795:432–437, 2019.

[83] Krzysztof Pachucki, Vojtěch Patkóš, and Vladimir A. Yerokhin. Recoil corrections with finite nuclear size in hydrogenic systems. Phys. Rev. A, 111(3):032820, 2025.

[84] Vojtěch Patkóš and Krzysztof Pachucki. Radiative corrections of the order $\alpha(Z\alpha)6$ for rotational states of two-body systems. Phys. Rev. A, 110(2):022813, 2024.

[85] Krzysztof Pachucki. Nuclear recoil correction to the hyperfine splitting in atomic systems. Phys. Rev. A, 106(2):022802, 2022.

[86] Gregory S. Adkins and Ulrich D. Jentschura. Relativistic and recoil corrections to light-fermion vacuum polarization for bound systems of spin-0, spin-1/2, and spin-1 particles. Phys. Rev. D, 113(5):056029, 2026.

[87] Gregory R Chambers, Fedor Manin, and Shmuel Weinberger. Quantitative null-homotopy and rational homotopy type. Geometric and Functional Analysis, 28(3):563–588, 2018.

[88] Richard D Canary. Ends of hyperbolic 3-manifolds. Journal of the American Mathematical Society, 6(1):1–35, 1993.

[89] Frank Quinn. Ends of maps. iii. dimensions 4 and 5. Journal of Differential Geometry, 17(3):503–521, 1982.

[90] Michael Francis Atiyah and Raoul Bott. The yang-mills equations over riemann surfaces. Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences, 308(1505):523–615, 1983.

[91] Cynthia Hog-Angeloni, Wolfgang Metzler, and Allan J Sieradski. Two-dimensional homotopy and combinatorial group theory, volume 197. Cambridge University Press, 1993.

[92] Robin A Swainson and Gordon WF Drake. A unified treatment of the non-relativistic and relativistic hydrogen atom ii: The green functions. Journal of Physics A: Mathematical and General, 24(1):95–120, 1991.

[93] Tatsumi Aoyama, Toichiro Kinoshita, and Makiko Nio. Theory of the anomalous magnetic moment of the electron. Atoms, 7(1):28, 2019.

[94] Axel Beyer, Lothar Maisenbacher, Arthur Matveev, Randolf Pohl, Ksenia Khabarova, Alexey Grinin, Tobias Lamour, Dylan C Yost, Theodor W Hänsch, Nikolai Kolachevsky, et al. The rydberg constant and proton size from atomic hydrogen. Science, 358(6359):79–85, 2017.

[95] Vladimir A Yerokhin, Zoltán Harman, and Christoph H Keitel. Two-loop electron self-energy for low nuclear charges. Physical Review Letters, 133(25):251803, 2024.

[96] Krzysztof Pachucki. Finite nuclear mass correction to the hyperfine splitting in hydrogenic systems. Physical Review A, 109(5):052822, 2024.

[97] Ulrich D Jentschura and Gregory S Adkins. Quantum Electrodynamics: Atoms, Lasers and Gravity. World Scientific, 2022.

[98] Owen Gwilliam and Brian R Williams. Higher kac-moody algebras and symmetries of holomorphic field theories. arXiv preprint arXiv:1810.06534, 2018.

[99] Owen Gwilliam and Kasia Rejzner. Relating nets and factorization algebras of observables: Free field theories: O. gwilliam, k. rejzner. Communications in Mathematical Physics, 373(1):107–174, 2020.

[100] Owen Gwilliam, Eugene Rabinovich, and Brian R Williams. Quantization of topological-holomorphic field theories: local aspects. arXiv preprint arXiv:2107.06734, 2021.

[101] Gerald V Dunne and Mithat Unsal. Resurgence and trans-series in quantum field theory: the $\mathbb{CP}^n$ model. Journal of high energy physics, 2012(11):170, 2012.

[102] Gerald V Dunne and Mithat Ünsal. New nonperturbative methods in quantum field theory: from large-n orbifold equivalence to bions and resurgence. Annual Review of Nuclear and Particle Science, 66(1):245–272, 2016.

[103] Maxim Kontsevich. Deformation quantization of algebraic varieties. Letters in Mathematical Physics, 56(3):271–294, 2001.

[104] Viktor S Fadin and LN Lipatov. Bfkl pomeron in the next-to-leading approximation. Physics Letters B, 429(1-2):127–134, 1998.

[105] Yuri V Kovchegov. Unitarization of the bfkl pomeron on a nucleus. Physical Review D, 61(7):074018, 2000.

[106] Edmond Iancu, Andrei Leonidov, and Larry D. McLerran. Nonlinear gluon evolution in the color glass condensate. 1. Nucl. Phys. A, 692:583–645, 2001.

[107] L Gårding. The theory of lacunas. In Hyperbolic Equations and Waves: Battelle Seattle 1968 Recontres, pages 13–21. Springer, 1970.

[108] Jean Écalle. Les algebres de fonctions resurgentes. 1981.

[109] David Sauzin. Nonlinear analysis with resurgent functions. In Annales scientifiques de l'École Normale Supérieure, volume 48, pages 667–702, 2015.

[110] Eric Delabaere, Hervé Dillinger, and Frédéric Pham. Exact semiclassical expansions for one-dimensional quantum oscillators. Journal of Mathematical Physics, 38(12):6126–6184, 1997.

[111] Nathan Seiberg. Emergent spacetime. In 23rd Solvay Conference in Physics: The Quantum Structure of Space and Time, pages 163–178, 1 2006.

[112] David Gross. The quantum structure of space and time: proceedings of the 23rd Solvay Conference on Physics, Brussels, Belgium, 1-3 December, 2005, volume 23. World Scientific, 2007.

[113] Lothar Maisenbacher. Precision Spectroscopy of the 2S-nP Transitions in Atomic Hydrogen. PhD thesis, Ludwig Maximilians Universität München, 2021.

[114] Cédric Lorcé, Hervé Moutarde, and Arkadiusz P. Trawiński. Revisiting the mechanical properties of the nucleon. Eur. Phys. J. C, 79(1):89, 2019.

[115] Dmitri E Kharzeev. Mass radius of the proton. Physical Review D, 104(5):054015, 2021.